

MAKE THE SYSTEM AGREE WITH ITSELF.

Integration is the mechanism that converts specialized engineering velocity into one launchable autonomy system?through reproducible behavior, traceable configuration, regression evidence, and clear ownership.

COMPANY MOMENT

Latitude is building a fully in-house ADAS platform for a 2027 Ford Universal Electric Vehicle launch, combining startup speed with Ford manufacturing and scaling discipline.

ACTUAL MANDATE

Turn cross-domain ambiguity into a bounded problem, coherent repair, regression evidence, traceable configuration, and ownership closure so partner teams can keep moving.

THE SIX COUPLED LAYERS

PLATFORM CONFIGURATION	identity + compatibility
COMPUTE + NETWORK	timing + resources
SENSORS + ACTUATION	calibration + state
AUTONOMY SOFTWARE	behavior + interfaces
VALIDATION EVIDENCE	reproduction + regression
OWNERSHIP + HANDOFF	decision + closure

FOUR CLOSURE MOVES

01 FIND THE FRACTURE

QUESTION

Where does observed behavior first diverge from expected system behavior?

DURABLE OUTPUT

Precise symptom, condition envelope, evidence gaps, candidate boundaries.

02 ESTABLISH THE INVARIANT

QUESTION

Which assumption must remain true across software, hardware, timing, data, calibration, or ownership?

DURABLE OUTPUT

Testable invariant and minimum reproducible condition.

03 RESTORE THE COUPLING

QUESTION

What code, configuration, interface, timing, calibration, or authority change makes the layers coherent?

DURABLE OUTPUT

Repair with explicit decision owner and adjacent-risk review.

04 MAKE RECURRENCE HARDER

QUESTION

Where should evidence, validation, compatibility, or process now fail fast?

DURABLE OUTPUT

Regression test, configuration check, observability, interface contract, or standard work.

MEASUREMENT CATEGORIES

Localization

Reproducibility

Blocker flow

Regression stability

Configuration
traceability

Recurrence
reduction

WHY THE EVIDENCE TRANSFERS

VAPE-JET	~1,000 fielded software-enabled robots	Cross-domain issue visibility, machine/factory/customer evidence, prioritization, handoffs, recurrence reduction.
ZEUSVU	Regulated autonomous drone operations; 100% safety record	Autonomous systems inside safety, environmental, sensing, mapping, stakeholder, and evidence constraints.
COMPUNETICS	Advanced electronics + high-reliability programs	Root cause, process controls, quality systems, hardware/software adjacency, technical translation.
AMAZON	24/7 launch environment at operating scale	Visibility, escalation, standard work, rapid prioritization, mechanism building under pressure.
DUDEWORTH	Python, TensorFlow concepts, agentic workflows	Software fluency, structured reasoning, workflow design, evaluation, governance, experimentation.

IMMEDIATE TECHNICAL CONTRIBUTION

INTEGRATION TRIAGE	PYTHON-ENABLED TOOLING	HARDWARE/SOFTWARE SYNTHESIS
CONFIGURATION TRACEABILITY	REGRESSION THINKING	CROSS-TEAM CLOSURE

90-DAY ENTRY LOGIC

DAYS 0-30 | LEARN

Set up the build/test environment. Learn repository structure, configuration management, telemetry, validation, compute constraints, and safety interfaces. Pair with domain owners. Close one bounded issue with a senior partner.

DAYS 31-60 | CONTRIBUTE

Own bounded integration work. Improve evidence capture and configuration comparison. Make blocker communication precise. Convert issue closure into targeted tests, controls, or interface documentation.

DAYS 61-90 | STRENGTHEN

Cluster recurring failure patterns. Propose reusable checks and tooling grounded in observed work. Apply the model to one new-platform or major-feature integration path.

EXECUTIVE INTERVIEW QUESTIONS

- 1 Where do responsibility gaps most often appear today?
- 2 What distinguishes a strong autonomy integration engineer after six months?
- 3 How are configuration lineage and release evidence represented across current platforms?
- 4 Which integration issues consume the most senior engineering attention, and why?
- 5 How does the team decide when a local fix is sufficient versus when the system contract must change?
- 6 What should become materially easier before the 2027 platform launch?

OPERATING PRINCIPLE

Ask precise questions. Pair with domain experts. Work directly in bounded problems. Document reasoning. Preserve ownership clarity. Let evidence determine the next experiment.